# STSM3764 GLM Interview: Model Answers

Each answer is written to hit the rubric's top band (5/5): accurate, detailed, with a GLM example. Say the **bold** parts; the rest is back-up detail. Examples to reuse: Challenger O-rings (logistic), potato yield vs nitrogen (gamma, log link), claim counts with exposure (Poisson, offset).

## 1. GLM vs OLS

**Why use a GLM instead of OLS?** OLS assumes a normal response with constant variance and a mean that is linear in the predictors. A GLM keeps the linear predictor but lets the response come from any exponential-family distribution and links the mean to the predictor through a function: $g(\mu) = \eta = X\beta$. This handles binary, count, and positive skewed data, and keeps predictions in the valid range (probabilities in [0,1], counts $\geq 0$).

**Main differences:**

- Distribution: normal (OLS) vs exponential family (GLM).
- Mean: $\mu = X\beta$ vs $g(\mu) = X\beta$.
- Variance: constant vs $\text{Var}(Y) = \varphi V(\mu)$, which depends on the mean.
- Estimation: least squares (closed form) vs maximum likelihood via iterative methods (IRLS/Fisher scoring).
- Inference: exact t/F tests vs asymptotic Wald/LR/deviance tests.

**Why always choose OLS?** OLS is a special case of a GLM (normal, identity link), so when the response is continuous, roughly symmetric, with constant variance, OLS is simpler, has exact inference, and a closed-form solution. **Why choose a GLM?** When the response is binary (Challenger O-ring failure), a count (claims), or positive and skewed with variance growing with the mean (potato yield, costs), where OLS would give impossible predictions and a wrong variance structure.

## 2. What do we use GLMs for?

Modelling the relationship between a response and explanatory variables when the response is non-normal: binary (logistic/probit), counts and rates (Poisson/negative binomial), positive continuous (gamma), and counts in contingency tables (loglinear models). Uses: explanation (effect sizes), prediction, and hypothesis testing.

## 3. Main components of a GLM

1. **Random component:** the distribution of $Y_i$, from the exponential family (e.g. Bernoulli for pass/fail O-rings).
2. **Systematic component (linear predictor):** $\eta_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}$.
3. **Link function:** $g(\mu_i) = \eta_i$, connecting the mean of the response to the linear predictor (e.g. logit for binary data).

## 4. Why does the exponential family matter?

If the distribution is in the exponential family, we get a **unified theory**: the same score equations, the same Fisher information, the same IRLS algorithm, the same asymptotic results, and a built-in mean-variance relationship, all expressed through one function $b(\theta)$. We also get the canonical link for free, and sufficient statistics. We only need to write down the family once, and any member plugs in.

## 5. How to check membership of the exponential family

Write the density in the form

$$f(y; \theta, \varphi) = \exp\{ [y\theta - b(\theta)] / a(\varphi) + c(y, \varphi) \}$$

The log density must split into: a term $y \cdot \theta(\mu)$, a term $b(\theta)$ depending only on the parameter, and a term $c(y, \varphi)$ depending only on y (and $\varphi$). Also, the **support must not depend on the parameter** (so Uniform(0, $\theta$)

is not in the family).

Example, Poisson: f = e^(−μ) μ^y / y! = exp{ y log μ − μ − log y! }, so θ = log μ, b(θ) = e^θ, a(φ) = 1.

**Mean-variance relationship:** E(Y) = b′(θ) and Var(Y) = a(φ) b″(θ) = φ V(μ). The variance is a fixed function of the mean: Poisson V = μ, binomial V = μ(1−μ), gamma V = μ², normal V = 1.

**How to find out if it is violated:** plot residuals (Pearson or DHARMa) against fitted values and look for a trend in spread; compare the residual deviance to its degrees of freedom (ratio much greater than 1 suggests overdispersion for Poisson/binomial); or compare empirical group variances to group means. Fixes: quasi-likelihood, negative binomial, or a different family.

## 6. Choosing the random component

Look at the **nature of the response**: binary → Bernoulli/binomial; counts → Poisson (or negative binomial if overdispersed); positive continuous skewed → gamma or inverse Gaussian; continuous symmetric → normal. Then check the mean-variance relationship against the data using plots, residuals and deviance.

**What is the gamma GLM used to model?** Positive, continuous, right-skewed data where the variance increases with the mean (Var ∝ μ², constant coefficient of variation), e.g. insurance claim sizes, waiting times, potato yields, rainfall.

**Positively skewed continuous data:** gamma (log or inverse link), inverse Gaussian (more extreme skew, V = μ³), or a normal GLM on log-transformed data. Gamma with a log link is the usual first choice because it keeps μ > 0 and effects are multiplicative. Choose by checking residual diagnostics and comparing AIC.

## 7. Canonical link function

**Best answer:** the canonical link is the one that sets the natural parameter equal to the linear predictor, θ = η. Practically: write the density in exponential form, take b(θ), compute **b′(θ) = μ**, then **invert** to get θ as a function of μ; that function is g.

Examples: Poisson b(θ) = e^θ, so μ = e^θ and g(μ) = log μ. Bernoulli b(θ) = log(1 + e^θ), so μ = e^θ/(1+e^θ) and g(μ) = logit(μ). Gamma gives the inverse link, 1/μ. Normal gives the identity.

**What does it mean when the link is canonical?** The natural parameter θ equals η = Xβ, and the **sufficient statistic for β is X'y** (Σ x_ij y_i), so the score equations simplify to X'(y − μ) = 0.

## 8. What do we want from a link function?

It must be **continuous, monotonic (strictly increasing or decreasing, so invertible and differentiable), and map the range of μ onto the whole real line**, so that any linear predictor gives a valid mean. Example: logit maps (0,1) to the real line; log maps (0,∞) to the real line.

### Given a brand new one-parameter distribution: BUILD then ESTIMATE

**Build:**

1. Write the density in exponential-family form and identify θ, b(θ), a(φ), c(y,φ). Check the support is free of the parameter.
2. Compute E(Y) = b′(θ) and Var(Y) = a(φ)b″(θ) to get the mean and variance function.
3. If the natural parameter is canonical: set b′(θ) = μ and invert to get θ = g(μ); this is the canonical link. Otherwise choose a link that is continuous, monotonic and maps the mean's range onto the real line.
4. Set g(μ_i) = x_i'β.

**Estimate:**

1. Write the log-likelihood ℓ(β) = Σ [y_i θ_i − b(θ_i)]/a(φ) + c.
2. Differentiate via the chain rule to get the score: U_j = Σ (y_i − μ_i) x_ij / [Var(Y_i) g′(μ_i)]. For a canonical link this is X'(y − μ)/φ.
3. Solve U(β) = 0 iteratively with Fisher scoring/IRLS; the inverse Fisher information gives the covariance of β.

## 9. Canonical vs non-canonical links

**Why canonical?** They make the score equations simple, the observed and expected information are equal (so Newton-Raphson and Fisher scoring coincide), there are sufficient statistics X'y, the log-likelihood is concave so the MLE is unique and convergence is well behaved, and there is a natural interpretation (logit gives log odds).

**Why non-canonical?** The canonical link may not respect the problem: the gamma inverse link can give negative means, whereas a log link guarantees positive means and gives multiplicative effects that are easier to interpret. The probit link has a natural latent-variable/dose-response interpretation. Complementary log-log suits asymmetric binary data. **Trade-off:** easier interpretation and better fit vs. losing the computational simplicity and guaranteed convergence.

## 10. Score equation and MLE

**Score equation:** $U(\beta) = \partial\ell/\partial\beta = \Sigma_i (y_i - \mu_i) x_i / [\varphi V(\mu_i) g'(\mu_i)] = 0$. For canonical links this reduces to $X'(y - \mu) = 0$. The score has mean zero at the true $\beta$.

**How is the MLE found?** There is no closed form, so use iteration: start with $\beta(0)$, then update $\beta(t+1) = \beta(t) + I^{-1} U(\beta(t))$ until convergence. This is equivalent to iteratively reweighted least squares: regress the working response z on X with weights W.

**Newton-Raphson vs Fisher scoring:** Newton-Raphson uses the **observed** information (minus the Hessian); Fisher scoring uses the **expected** information $E[-\partial^2\ell/\partial\beta\partial\beta']$. Fisher scoring is more stable and gives IRLS; with canonical links they are identical.

**What is the Taylor expansion used for?** To linearise the score function around the current estimate (first-order expansion of $U(\beta) = 0$), which gives the iterative update. It is also used in the asymptotic argument that $\beta$ is approximately normal.

**What does the inverse Fisher information give us?** The **asymptotic variance-covariance matrix of $\beta$**, so $\beta \sim N(\beta, I(\beta)^{-1})$. Square roots of the diagonal give standard errors for Wald tests and confidence intervals.

## 11. Interpreting slope estimates

| Link | Model | Interpretation of β1 |
|---|---|---|
| Identity | $\mu = \beta0 + \beta1\ x$ | A one-unit increase in x changes the mean by β1 units (additive). |
| Log (β = 0.2) | $\log \mu = \beta0 + \beta1\ x$ | A one-unit increase in x multiplies the mean by e^0.2 ≈ 1.22, i.e. about a 22% increase. |
| Logit | $\log[p/(1-p)] = \beta0 + \beta1\ x$ | A one-unit increase in x multiplies the **odds** by e^β1 (changes the log odds by β1). |
| Inverse | $1/\mu = \beta0 + \beta1\ x$ | A one-unit increase in x changes the reciprocal of the mean by β1; a positive β1 means the mean decreases. Not directly intuitive. |
| Probit | $\Phi^{-1}(p) = \beta0 + \beta1\ x$ | A one-unit increase in x changes the **z-score** (standard normal quantile) of the probability by β1. |

## 12. Challenger example

Binary response (O-ring thermal distress/failure), temperature as predictor, logistic regression. The temperature coefficient is **negative and significant**: as temperature falls, the odds of O-ring failure increase, with the odds multiplied by e^β per degree. Launch day was about 31°F, far below the observed range (about 53–81°F), so predicting there is **extrapolation** and the fitted probability of failure is very high. From the output I would also report the standard error and Wald z/p-value, the residual vs null deviance (improvement in fit), and AIC. The lesson: the data signalled the risk, and a GLM quantifies it, while warning about extrapolation.

## 13. Goodness of fit and testing

**Goodness of fit of a single GLM:** compare the residual deviance $D = 2[\ell(\text{saturated}) - \ell(\text{model})]$ to a $\chi^2$ with $n - p$ degrees of freedom (valid for grouped binomial/large-count Poisson data, **not** for ungrouped binary data). Also use Pearson $\chi^2$, and DHARMa residuals or residual plots.

**What can Wald tests do?** Test a single parameter ($z = \beta/SE(\beta)$, or $z^2 \sim \chi^2(1)$); test several parameters jointly, via $W = (C\beta - c)'[C\, I^{-1}\, C']^{-1}(C\beta - c) \sim \chi^2(q)$; be used for goodness of fit (e.g. testing that the extra parameters of a saturated model are zero); and build confidence intervals.

**Testing multiple parameters simultaneously:** joint Wald test, or likelihood ratio/deviance test comparing nested models, $2[\ell(\text{full}) - \ell(\text{reduced})] \sim \chi^2(q)$ where q is the number of extra parameters.

**Wald vs LR advantage:** Wald only needs the **full model to be fitted** (no refit), is cheap, and gives parameter-wise output directly. **LR is more reliable**, especially for small samples or parameters near the boundary, and is invariant to reparametrisation; Wald can fail with large coefficients (Hauck-Donner effect, e.g. perfect separation).

**Very small residual deviance:** the model fits about as well as the saturated model, which means good fit, **but** it could also be **overfitting** (too many parameters), or underdispersion. Compare to degrees of freedom; a deviance far below df is suspicious.

**Wald vs deviance vs LR:**

- **Wald:** uses $\beta$ and its estimated variance at the MLE; asymptotically $\chi^2$.
- **LR:** compares maximised log-likelihoods of two nested models: $2(\ell 1 - \ell 0)$.
- **Deviance test:** a special case of LR where the reduced model is compared with the saturated model; the difference in deviances between two nested models equals the LR statistic ($\varphi = 1$).
- All three are asymptotically equivalent; LR/deviance are generally preferred.

## 14. Dispersion parameter

$\varphi$ scales the variance: $\text{Var}(Y) = \varphi\, V(\mu)$. For normal and gamma it must be estimated ($\varphi = \sigma^2$ for normal); for Poisson and binomial it is fixed at 1. It measures how much more (overdispersion, $\varphi > 1$) or less (underdispersion, $\varphi < 1$) variable the data are than the model predicts. Estimated by Pearson $\chi^2 / (n - p)$.

**The problem:** for Poisson/binomial the assumption $\varphi = 1$ is often false in real data (overdispersion from heterogeneity or clustering), which makes **standard errors too small**, so tests are too liberal. Remedies: quasi-Poisson/quasi-binomial (scale standard errors by $\sqrt{\hat{\varphi}}$), negative binomial, or random-effects models.

## 15. DHARMa residuals

Standard residuals for discrete data are not normal, so plots are hard to read. **DHARMa simulates** many responses from the fitted model, and for each observation computes the **quantile of the observed value within its simulated distribution** (randomised for discrete data). If the model is correct, the scaled residuals are **Uniform(0,1)**. We use them for QQ plots (uniformity), residuals vs predicted (patterns), and formal tests for **uniformity, dispersion, zero-inflation and outliers**.

## 16. Dose-response and probit

**Dose-response model:** models the probability of a response (e.g. death, failure) as a function of dose or exposure, $p(d) = F(\beta 0 + \beta 1\, d)$ with F a CDF (logit, probit, cloglog). Often used to find the **LD50/ED50** $= -\beta 0/\beta 1$ (dose where $p = 0.5$), often with log-dose.

**Probit coefficients:** each unit increase in dose increases the standard normal quantile (z-score) of the response probability by $\beta 1$. This corresponds to the **tolerance interpretation**: each subject has a normally distributed tolerance threshold, and responds when the dose exceeds it. The effect on p itself depends on the dose ($\varphi(\eta)\beta 1$ is the slope).

## 17. Contingency tables and loglinear models

**Applying a GLM to a contingency table:** treat the cell counts as **Poisson** (or multinomial/product multinomial, which gives the same likelihood-based inference when the fixed margins are included) with a **log link**: $\log \mu_{ij} = \lambda + \lambda_{i(A)} + \lambda_{j(B)} + \lambda_{ij(AB)}$.

**Main purpose:** to model **associations and dependence structure between categorical variables** (not a response vs predictors), and to test which interactions are needed.

**Interpreting parameters:** main effects reflect the marginal (relative) frequencies of the categories; **interaction terms are log odds ratios**, so exp(λ_ij(AB)) is a multiplicative departure from independence. In a 2×2 table the single interaction equals the log odds ratio.

**Offset:** a term with a **fixed coefficient of 1** added to the linear predictor: log μ_i = log(t_i) + x_i'β, so that we are modelling the **rate** μ_i/t_i. For count data, t_i is the exposure (time, population, area), because counts depend on the length of exposure. For contingency tables the offset represents **fixed margins or totals** (a conditioned total).

**Testing independence:** fit the independence model (main effects only) log μ_ij = λ + λ_i + λ_j, and compare its **residual deviance (or Pearson $\chi^2$) to $\chi^2$ with (I−1)(J−1) df**. A small deviance (large p-value) means independence is adequate. Equivalently, an LR/deviance test of the interaction term against the independence model: if the interaction is significant, the variables are **associated**. This is the same as the classical $\chi^2$ test of independence.

## 18. Explain a GLM to a Grade 12 student using functions

**Say it in four steps (about 30 seconds each):**

1. **Start with a function they know.** A straight line, y = mx + c, takes an input (nitrogen applied) and gives an output (potato yield). That is ordinary linear regression: the output can be any number, positive or negative.
2. **Show where the line breaks.** Some outputs can't be any number: a probability must stay between 0 and 1, a count can't be negative, a yield must be positive. A straight line eventually crosses those limits and gives nonsense.
3. **Wrap the line inside another function.** Keep the line, call it η = β0 + β1x, but feed it into a function that forces valid outputs: μ = e^η (an exponential function, always positive) or p = 1/(1 + e^−η) (the S-shaped logistic curve, always between 0 and 1). Then, **the link function is just the inverse function**: log undoes the exponential, logit undoes the logistic. That is exactly "inverse functions" from their course.
4. **Admit that real data scatter around the curve.** The curve gives the average outcome; the data vary around it, and how much they vary depends on the situation: yes/no outcomes (Bernoulli), counts (Poisson), skewed positive amounts (gamma). That choice of scatter is the **random component**.

**The three parts in function language:** the straight line inside (linear predictor), the function wrapped around it (inverse link), and the shape of the scatter (distribution).

**Why the exponential is a good example:** with μ = e^(β0 + β1x), every one-unit increase in x **multiplies** the output by the same factor e^β1, which is exponential growth. So β1 = 0.2 means about a 22% increase per unit.

**How do we pick the curve?** Instead of minimising squared distances, we choose the β values that make the observed data most likely under the curve (maximum likelihood). A computer does this by trial and improvement (iteration).

**One-line summary to finish:** "A GLM is a straight line put inside a function so that the answers always make sense, plus an honest description of how the data scatter around that curve."

## 19. Steps for testing goodness of fit of a GLM

1. **State the hypotheses.** H0: the model fits adequately (it is as good as the saturated model). H1: it does not fit (lack of fit).
2. **Check the test is valid.** The $\chi^2$ approximation needs grouped binomial data or reasonably large Poisson counts (expected counts of about 5 or more). For **ungrouped binary data** (e.g. one row per O-ring) do not use it; use the Hosmer-Lemeshow test or DHARMa instead.
3. **Fit the model and get the residual deviance D** (and the Pearson $\chi^2$), with residual degrees of freedom n − p.
4. **Compare to $\chi^2$ with n − p df.** p-value = P($\chi^2$(n−p) ≥ D). A **large p-value means no evidence of lack of fit**; a small p-value means the model fits poorly.

5 **Quick rule of thumb:** D / df should be close to 1. Much larger than 1 points to overdispersion, a missing predictor or a wrong link; much smaller than 1 points to underdispersion or overfitting.

6 **Look at the residuals.** Pearson/deviance residuals vs fitted values, a DHARMa QQ plot, and tests for dispersion, outliers and zero-inflation. Check influential points (Cook's distance).

7 **If the fit is poor, fix it:** add missing terms, transformations or interactions; try another link; allow for overdispersion (quasi-likelihood or negative binomial); or consider zero-inflation. Then compare candidate models with the LR test (nested) or AIC (non-nested).

8 **Conclude in context.** State the test statistic, df and p-value, and what it means for the example (e.g. "the residual deviance of 20.3 on 21 df gives p = 0.5, so no evidence of lack of fit").

## 20. Deviance tests in R

**Idea:** deviance $D = 2[\ell(\text{saturated}) - \ell(\text{model})]$. A **difference in deviances** between two nested models equals the LR statistic and is compared to $\chi^2$ with df equal to the number of extra parameters. For Poisson and binomial ($\varphi$ fixed at 1) use $\chi^2$; for gamma, normal or quasi families ($\varphi$ estimated) use an **F test**.

```
fit0 <- glm(y ~ 1,       family = poisson, data = d)   # null model
fit1 <- glm(y ~ x1,      family = poisson, data = d)
fit2 <- glm(y ~ x1 + x2, family = poisson, data = d)
summary(fit2)    # shows null deviance, residual deviance, df, AIC

# 1. Goodness of fit: residual deviance vs chi-square (small p = lack of fit)
pchisq(deviance(fit2), df.residual(fit2), lower.tail = FALSE)

# 2. Overall model test: is the model better than null?
pchisq(fit2$null.deviance - fit2$deviance,
       fit2$df.null - fit2$df.residual, lower.tail = FALSE)

# 3. Nested model comparison (deviance / LR test), by hand and with anova()
dev_diff <- deviance(fit1) - deviance(fit2)
pchisq(dev_diff, df.residual(fit1) - df.residual(fit2), lower.tail = FALSE)
anova(fit1, fit2, test = "Chisq")

# 4. Term-by-term tests
anova(fit2, test = "Chisq")    # sequential (order matters)
drop1(fit2, test = "Chisq")    # drop each term from the full model

# 5. When dispersion is estimated (gamma, quasi-): use an F test
anova(fit1, fit2, test = "F")

# 6. Dispersion check and residual diagnostics
sum(residuals(fit2, type = "pearson")^2) / df.residual(fit2)   # near 1 is good
library(DHARMa)
sim <- simulateResiduals(fit2); plot(sim); testDispersion(sim)

# 7. Compare non-nested models
AIC(fit1, fit2)

# 8. Independence test in a loglinear model (I x J table, counts in 'n')
ind <- glm(n ~ A + B, family = poisson, data = tab)
pchisq(deviance(ind), df.residual(ind), lower.tail = FALSE)  # df = (I-1)(J-1)
```

**How to read the** `anova()` **output:** columns are Resid. Df, Resid. Dev, Df, Deviance and Pr(>Chi). "Deviance" is the drop in residual deviance from adding the term(s), "Df" is the number of parameters added, and a small Pr(>Chi) means the extra terms significantly improve the fit. Say the result in words: "adding x2 reduces the deviance by 9.4 on 1 df (p = 0.002), so x2 is needed."

**Common points to mention:** the models must be **nested** and fitted to the **same data**; `anova()` on one model is **sequential** so term order matters, whereas `drop1()` is not; the deviance difference is generally more reliable than the Wald test from `summary()`.

## Closing line for any question

If stuck: state the **definition**, say **why it matters**, give **one GLM example** (Challenger logistic, Poisson counts with offset, or gamma log-link for skewed yields). That structure alone secures a 4-5 under the rubric.